01 Optimising Accessible 3D Pipelines for In Situ Osteological Recovery: Methodological Bottlenecks and FLOS Integration

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The in situ 3D documentation of human remains presents operational challenges due to morphological complexity, joint articulation, and severe undercuts inherent in biological materials. This research presents a case study evaluating a multimodal 3D acquisition and processing pipeline—firmly grounded in FLOS (Free, Libre, and Open Source) tools and FAIR principles—designed to optimize the digital recovery of osteological evidence in funerary contexts. The experiment has been conducted on a simulated archaeological burial, relying on remains from the Mulazzano archaeological site, dating back to the 6th century AD. The study tests a modified Structure from Motion (SfM) field protocol paired with sub-millimeter close-range structured light acquisition. Ongoing analysis evaluates potential bottlenecks across both standard capture workflows and subsequent computational pipelines, specifically addressing material interference and critical occlusion zones generated by the geometric complexity of skeletal remains. By systematically mapping these criticalities within a controlled environment, the study aims to refine FLOS-based protocols and establish a robust methodological baseline for data reproducibility, remote morphological analysis, and the ethical, non-invasive management of human remains within Virtual Anthropology.